

FILTER DESIGN TEST CASE

Notch Filter - WiFi 5 GHz for Automotive V2X Communication
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Design Center:	Newbury Park Design Center
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Priority:	P3 - Medium

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Notch Filter filter targeting WiFi 5 GHz for Automotive V2X Communication. The filter is designed on ScAlN on Si substrate with a target center frequency of 3709.2 MHz and bandwidth of 274.3 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Automotive V2X Communication module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Notch Filter
- Frequency Band: WiFi 5 GHz
- Application: Automotive V2X Communication
- Substrate: ScAlN on Si
- Package: WLP (Wafer Level Package)
- Design Tool: CST Studio Suite 2024
- Design Center: Newbury Park Design Center

This specification applies to all variants of the Notch Filter design targeting WiFi 5 GHz. Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3709.2 MHz
Bandwidth (3 dB):	274.3 MHz
Insertion Loss Target:	≤ 1.5 dB
Return Loss Target:	≥ 19.0 dB
Out-of-Band Rejection:	≥ 33.0 dB
Isolation (Duplexer):	≥ 28.0 dB
Q Factor:	9812
Temperature Coefficient:	-14.9 ppm/°C
Die Size:	0.9 x 1.3 mm
Wafer Size:	8-inch
Number of Resonators:	4
Electrode Material:	Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open CST Studio Suite 2024 and load the Notch Filter template project.
2. Define the substrate stack: ScAlN on Si with specified layer thicknesses.
3. Set design targets: center freq 3709.2 MHz, BW 274.3 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.5 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (WLP (Wafer Level Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, ScAlN on Si).
10. Perform on-wafer probing using Keysight E5080B ENA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 11128 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.5 dB
- Return loss in passband shall be >= 19.0 dB
- Out-of-band rejection at +/- 411 MHz offset >= 33.0 dB
- Group delay variation across passband shall be <= 10 ns
- Temperature drift over -40°C to +85°C shall be <= 6.9 MHz
- Power handling shall be >= +31 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 97%

6. TEST RESULTS

Measurement Date: 2024-11-12
Devices Tested: 43
Insertion Loss (meas): 1.19 dB
Return Loss (meas): 20.9 dB
Rejection (meas): 36.4 dB
Isolation (meas): 32.7 dB
Q Factor (meas): 9812
Group Delay Variation: 3.3 ns
Power Handling: +26 dBm
Die Yield: 91.2%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The Notch Filter design demonstrated excellent performance for WiFi 5 GHz.
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Simulation-to-measurement correlation was within 3% for all key parameters.
- e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

Author: Dr. Raj Patel Date: 2024-11-12
Reviewer: Dr. Anya Krishnamurthy Date: 2024-11-14
Approver: Dr. Karen Mitchell Date: 2024-11-22

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